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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE(24AD3107)

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TITLE OF THE PROJECT	Sentiment Analysis of Tweets	
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Abstract

In the current era of digital communication, social media platforms such as Twitter (X) have become one of the most influential sources of real-time public opinion. Millions of users express their thoughts, emotions, and reactions on a wide range of subjects, including political events, product launches, sports, entertainment, and social issues, generating an enormous and continuously growing volume of unstructured textual data. Manually analyzing this data to understand public sentiment is impractical due to its sheer scale and the rapid pace at which new content is generated. This creates a strong need for automated systems capable of extracting, processing, and interpreting sentiment from such data efficiently and accurately. This project addresses that need by presenting a sentiment analysis system that collects tweets related to a chosen trending topic and classifies them into positive, negative, or neutral categories using Natural Language Processing (NLP) techniques.

The overall methodology of the project begins with the collection of tweet data using the Twitter API or a suitable pre-existing dataset, focusing on a specific trending topic, hashtag, or keyword to ensure the relevance and timeliness of the analysis. Since raw tweets are inherently noisy and unstructured, an extensive preprocessing pipeline is applied before any meaningful analysis can take place. This includes tokenization, conversion to lowercase, removal of stop words, stemming and lemmatization, as well as the careful handling of hashtags, user mentions, emojis, special characters, and embedded URLs, all of which are common in tweet text but can interfere with accurate sentiment detection if not properly managed. Once the text has been cleaned and normalized, meaningful features are extracted using techniques such as Bag-of-Words, Term Frequency-Inverse Document Frequency (TF-IDF), or dense word embeddings that capture semantic relationships between words.

Following feature extraction, sentiment classification is performed using either traditional machine learning algorithms, such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression, or more advanced deep learning architectures, such as Long Short-Term Memory (LSTM) networks and transformer-based models like BERT, depending on the scope, computational resources, and accuracy requirements of the project. Each of these approaches offers a different trade-off between simplicity, interpretability, training time, and classification accuracy, and the project explores these trade-offs to identify the most suitable model for the task at hand. The trained model is then used to assign a sentiment label of positive, negative, or neutral to each tweet, effectively transforming raw, unstructured text into structured, actionable insight.

To make the results more interpretable and useful, the classified sentiments are visualized through various graphical representations, including pie charts, bar graphs, sentiment trend lines over time, and word clouds that highlight the most frequently occurring terms within each sentiment category. These visualizations help reveal overall public sentiment trends and patterns, making it easier to understand how people are reacting to a specific topic at a given point in time, and how that sentiment may shift as new events unfold. To ensure the reliability and correctness of the classification system, its performance is rigorously evaluated using standard metrics such as accuracy, precision, recall, and F1-score, with results compared across the different models tested during the course of the project.

Overall, this project demonstrates the practical application of Natural Language Processing and machine learning techniques in extracting meaningful, actionable insights from large volumes of unstructured social media data. Beyond its academic value, the system has significant real-world applicability across multiple domains, including brand monitoring and reputation management, political sentiment and election analysis, market research and consumer behavior studies, customer feedback analysis, and crisis or disaster response management, where understanding public mood in real time can support faster and more informed decision-making. By combining data collection, text preprocessing, feature engineering, model training, and result visualization into a single end-to-end pipeline, this project provides a comprehensive and practical demonstration of how sentiment analysis can be applied to real-world social media data.

Signature of the Guide